On Magnolia Square: Website Development Proposal

Neo Alabastro

October 5, 2023

Summary

1 Introduction

This proposal, geared toward those without a technical background, presents services, tools, and strategies to aid On Magnolia Square's members who are interfacing with the website's aesthetic architecture, content management, and technical stability, considering the lowest common denominator of technical literacy. Each proposed service or tool consists of general information and financial costs. This proposal also describes team roles needed to complete a full transition from the current website to a new one, and concludes with a sequential phase roadmap. This introduction section provides purposes and explanations for the rhymes and reason for this proposal's creation.

1.1 Problem

For the On Magnolia Square (OMS) team, the current OMS website has an obtuse and unwieldy website management services and tools required for its maintenance. For the reader, the OMS website lacks important security fixes and a cohesive user interface and experience, consequently reducing readership and public interest.

1.2 Goal

Develop and deploy a new OMS website that will not only repair the current website's issues of security, aesthetic, and content management, but will also serve as a stable foundation on which current and future OMS members, regardless of their technical ability, may be able to create, maintain, update, and manage self-published articles, photos, and content.

1.3 Solution

The solution is to employ modern services and tools to achieve the aforementioned goal, while also providing proper documentation for each part of the process for future OMS members to reference.

1.4 Purpose

Throughout this organization's history, no members have indulged in an in-depth proposition to change the OMS website architecture. Changes have manifested, but have culminated into the website as it is today. It is functional, but given the problems in §1.1, its potential to be a cornerstone of the New York University Shanghai (NYUSH) community is unnecessarily limited, even with the many shades of undergraduate talent intrinsic to our academic atmosphere. Completely online publications like OMS are generally very flexible in their execution, so I believe it is important to reify a logical game plan present and future members can utilize to focus on guaranteeing purposeful journalism.

2 Terminology

For the purpose of speaking the same language, this section introduces terminology required to understand the rest of this proposal.

2.1 Technical Terms

Understanding the construction of a new website requires familiarity with website terminology. Please remember that this reading is geared toward those with non-technical backgrounds, and does not require intimate knowledge of web technologies to understand.

The Internet is analogous to a highway: a series of roads that connect distinct locations, like your computer or mobile device, to other distinct locations, like websites and remote servers. Traveling on the highway is typically done using a web browser. Similar to a personal chauffer, a web browser prompts you for an address. Where do you want to go? That address is called the **Uniform Resource Locator (URL)**. These addresses are composed of several parts, but what matters for our purposes is the **domain name**. For instance, take this URL:

https://www.google.com/search?q=nyu+shanghai

The domain name is *google.com*. Other examples of domain names are punahou.edu, minecraft.net, or onmagnoliasquare.com. Another term that often comes up is with the term domain name is **Top Level Domain (TLD)**, which just means the two to three letter segment after the dot. For example, edu, net, and com are all TLDs. For our domain,

https://www.onmagnoliasquare.com

The domain name is *onmagnoliasquare.com*. The TLD is *com*.

A **domain name registrar** provides domain name registration and leasing services. There are many domain name registrars, like GoDaddy, Namecheap, or Amazon Route 53. Domain names are leased on a yearly basis. Registrars often give the option to pay for two, five, or even 10 years in advance. Our current domain name registrar for OMS is GoDaddy.

Sometimes, domain name registrars offer **website hosting** services. For example, if a website was a building, some domain name registrars do not only let you register an address for your website, but they also lease a plot of land you can build your building on; they lease a physical space people can actually see and visit.

Now, imagine that the web browser is your personal chauffer once again. They take your provided URL address, https://onmagnoliasquare.com, and drive you to the OMS website. Before unlocking the doors and letting you off at the location, they check in with the security at the gates. Your chauffer wants to validate that the address you provided and your current location is indeed the place that you want to be. In other words, you gave your chauffer the address for the OMS website and now that you have arrived, they want to ensure that yes, this location is actually the OMS website, or no, this is not the OMS website, and the address you provided is wrongly associated with the current location. This validation is done with a **certificate**[1], a cryptographic electronic verification validating true ownership of a domain issued by a type of organization called a **Certificate Authority (CA)**. Your chauffer rolls down the window so location security can hand over their certificate for your chauffer to inspect. Upon inspection, your chauffer will either let you off or freak out. In the case that they freak out, it is most likely for this reason: the certificate from location security is a fraud! It is self-signed, meaning it was not issued by a valid CA. They turn to you and say: 'Hey, there's an issue: your personal security may be at risk if I let you off at this location. After inspecting this location's certificate, it appears to be self-signed. Do you still want me to drop you off here?'

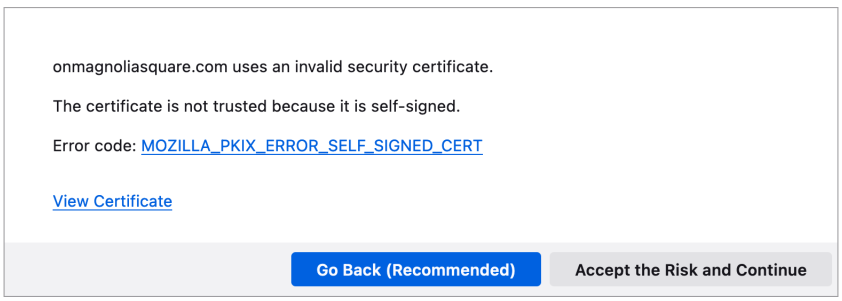

Figure 1: A self-signed certificate warning in the Firefox web browser.

It is up to your discretion to exit the vehicle. Such a warning would scare off many people and tell their chauffer to bring them back home.

All the terms I have described thus far refer to the **backend**. It is infrastructure that our typical website reader is unaware of, nor needs to know anything about. Maintaining an online presence using a website is like hosting a theatrical play. You want an audience to come and watch. They do not care about the stagehands or the prop designers or even the words on the script or anything else happening off-stage. They care about the performance and the plot. They care about the **frontend**.

The frontend is typically associated with the term **User Interface and User Experience (UI/UX)**. It simply describes the qualities of the way a user of something interfaces with it, and the experience that accompanies using that interface.

For example, a wooden pencil has a simple user interface: a slightly long, cylindrical instrument beginning at a short rubbery stub, and terminating at a fine, cone-shaped tip. A pencil also has a non-problematic user experience: users can hold it in any which way they please without impairment to handwriting or

[1] In technical terms, this is called an SSL X.509 Domain Validated Certificate.

drawing technique. These two characteristics of its UI/UX grant the user the ability to apply the graphite tip to a flat and markable medium in a sufficient and comfortable manner, such that any hand movements required to produce a writing or drawing can be done effectively, efficiently, and easily.

The content of many journals or newspaper websites, for example, the articles you read on JSTOR or The New York Times is created, updated, and deleted using a **content management system (CMS)**. A CMS interfaces between the frontend and the backend. It takes information from the backend, like articles or images, and sends them to be displayed at the frontend. A user who wants to update content on a website uses a CMS. A CMS is a tool developed for publishers, editors, and content creators, and so the UI/UX of a CMS is accessible for those genres of work.

A CMS can be imagined as the pencil in the UI/UX example. It interfaces between your brain and its ideas, and with the physical medium on which the idea will be presented. A CMS with a good UI/UX should feel fluent and fast, similar to how a pencil with a good UI/UX should feel comfortable and familiar.

2.2 Qualitative Terms

Each proposed backend and frontend service is chosen for its demonstration of good qualities in three categories of traits:

1. Accessible
2. Robust
3. Future-proof

Because of these traits, these services or tools can be used in the hands of a novice or experienced OMS member, with varying levels of technical ability, to achieve a satisfying outcome regardless of their goal.

Accessible

Accessible backend services boast a user friendly UI/UX. The OMS members operating the website in the future must have a successful and satisfying experience. Malfunction should not result from bad design. Accessibility also means that technical literature on the service is easily accessible from online documentation, online forms, or tech support.

Robust

A robust service is one that must be able to be kicked and tossed to the ground without compromise to its dependability. It is also mistake agnostic – an ignorant mistake or a well conceived blunder by an OMS member using the service will not hinder the service's reliability to display the website to our audience. In the case where an issue occurs, information to fix it is readily accessible.

Robust services also beget high standards of security, in the form of accessible user access management systems, multi-factor authentication, or reliable customer support.

Future-proof

Regardless of a service's accessibility and robustness, if it will not exist in the future, it is not employable in the present. Future members members must be able to access and use the service. I have chosen services created by strong and reputable companies in their respective technical circles, such that if a breaking change for their customers were to ever emerge down the line, a warning or notice will have been announced several months beforehand. This gives time for members to explore other service solutions.

3 Infrastructure

This section will discuss the current backend and frontend services and how they are put together. Each section will have a diagram depicting the system that creates the OMS website, comprised of the services involved and their interactions with each other. Arrows in each section depict the flow of content and information.

3.1 Current Infrastructure Plan

Our current infrastructure uses GoDaddy as an End-To-End (E2E) solution. An E2E system has a solution for every part of the process of a deployment pipeline. For us, GoDaddy accommodates the entire website publishing spectrum, beginning from the writing and drafting, to publishing and design, and concluding at the eyes of the audience. Every service is consolidated under GoDaddy making it a one-stop-shop for everything, begetting convenience for simple content management; however, this approach is prone to vendor lock-in and rigidity, which both violate the accessibility, robustness, and future-proofing qualities covered in §2.2.

I suppose E2E infrastructure was implemented because the GoDaddy route at the inception of OMS (previously On Century Avenue), was a conventionally tried-and-true, albeit severely sub-par, method of website creation.

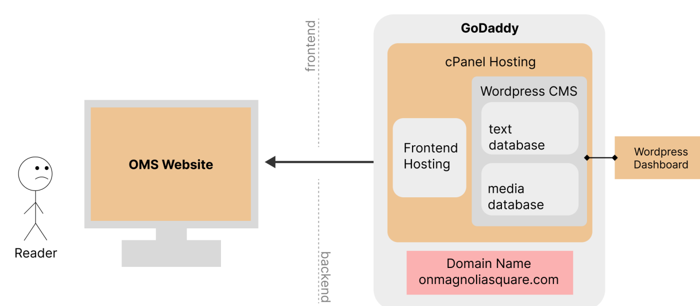

Figure 2: An End-To-End (E2E) infrastructure plan.

GoDaddy provides hosting under service *cPanel*. cPanel hosts our Wordpress CMS, its accompanying text and media database, as well as the frontend. Content flows from the Wordpress dashboard, the place where publishers input text and images, into a text and media database, and then onto the frontend where readers can view the content.

The current infrastructure uses archaic technology. Not only is GoDaddy frowned upon, but cPanel is as well. cPanel is a website hosting service that currently hosts our Wordpress CMS and frontend. Due to cPanel's age, its UI/UX is sluggish and frustrating to use.

Wordpress is an old and bloated CMS. Although it may be robust (it has stood the test of time), it is slow and easy to mangle.

3.2 Proposed Infrastructure Plan

My proposed infrastructure uses a Best-Of-Breed (BOB) solution. A BOB system is like composing a three-piece suit from materials sourced from the best silk and cotton mills; every component that constitutes the website is the crème de la crème, or for our purposes, the ones that suit our requirements the best.

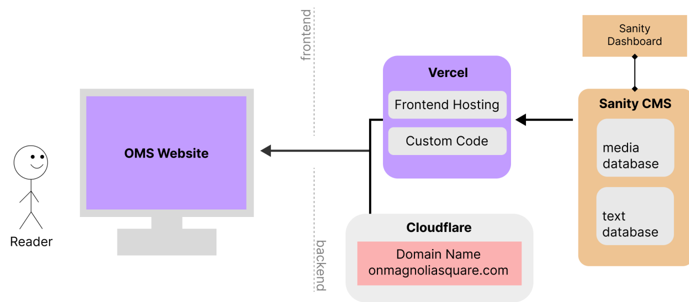

Figure 3: A best-of-breed infrastructure.

Similar to the GoDaddy E2E solution, content flows from the Sanity CMS dashboard, through to the CMS host, then to the frontend host, Vercel, which is then displayed to the reader. Multiple services interacting with one another is called decoupling, and doing this ensures flexibility. Content is content and services are services. There is no blending of the two.

Using a BOB solution also enables easier diagnosis of problems that may arise in the future, because the components of the system are partitioned into discrete units with single responsibilities rather than consolidated into one box.

3.3 Ditch or Fix?

Rather than fixing the current infrastructure and ensuring that it maintains longevity into the future, I feel that it is better to ditch it, because the time spent attempting to fix and then ensuring future-proofing is better spent on creating new infrastructure from the bottom up. Employing modern digital solutions to publish information is always a sprinter's marathon: to achieve long term success, you must always be the quickest to adopt new tech. Therefore, our publication requires rocket boosters strapped to its back to stay relevant. That cannot be done with our current infrastructure.

4 Proposed Services and Tools

This section proposes new and better backend and frontend services that will enable the smooth operation of maintaining the website's infrastructure, development, and content. Each proposed service also includes the issue that they fix.

4.1 Cloudflare

Cloudflare is a company that services and maintains digital cloud infrastructure across the globe. Cloudflare has good qualities in the three trait categories. Cloudflare is accessible: the UI/UX is simple and clear. Access to different settings and functions is clearly labeled. In the event that something may be unclear, their online documentation[2] and blog[3] is accessible and digestible. The services they provide are robust, dependable, and future-proof because a sizable majority of the internet flows through their worldwide data centers. Simply put, cloud security and internet infrastructure is their domain. Because of this, they provide backend service products we're particularly interested in: their registrar and photo storage.

Domain Registrar

In the technical community, using GoDaddy is a sin; any other domain registrar is better than GoDaddy. I chose Cloudflare's product, named *Registrar*, because of its metric gathering and ease of use. Cloudflare is one of several domain registrars that offer a free certificate[4] for any domain on their platform, along with enhanced customizability for certificate options and security[5]. Cloudflare's analytics and metrics aggregation is also top-tier.

Please note that readers who view the website will be unaware of this registrar switch – visiting the OMS URL will still display the same website content before the switch. It does not affect the frontend.

Image Storage and Hosting

Price Breakdown

4.2 Sanity

Sanity is a *headless* CMS. Our current CMS, Wordpress, is not a headless CMS, because it has a frontend component that shows the content from the CMS to readers (§3.1, Figure 2). A headless CMS gives more flexibility to designers and web developers who want granular control over the UI/UX of the frontend product; the way content is displayed and the content itself are decoupled entities.

[2]https://developers.cloudflare.com
[3]https://blog.cloudflare.com
[4]https://cloudflare.com/application-services/products/ssl/
[5]https://developers.cloudflare.com/ssl/get-started

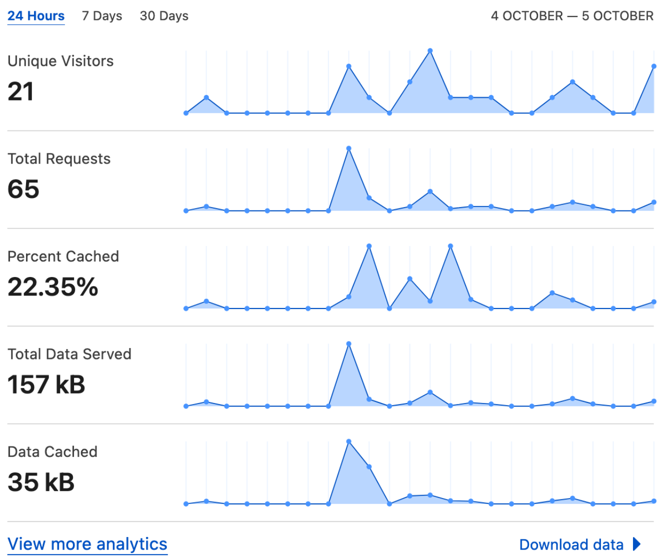

Figure 4: A view of the Cloudflare Registrar domain metrics dashboard.

Features

A headless CMS like Sanity may seem to separate two inseparable components of creating a website, that is publishing content and UI/UX. After all, UI/UX is a medium for the content to be displayed.

Since content and UI/UX design are separated, writers and publishers will not need to treat design as an afterthought. Sanity translates the results of the collaboration between web developers and UI/UX designers into a domain in which non-technical publishers are able to design a page, regardless of the page's content, to their liking. Designers create the design of components for the web developers to implement into real code. Sanity consumes this component code and prepares it for writers and publishers to incorporate into their content on the frontend. Publishers then combine both written content and custom components to create articles or web pages that are published on the frontend. This approach can satisfy the customizability needs of the website to have a more coherent UI/UX design language, as well as future-proofing since components created now can be used well into the future.

4.3 Vercel

Vercel provides website hosting, named *Vercel Frontend Cloud*, for frontend projects. Vercel manages the nitty-gritty of hosting the frontend of a website, and lets its users concentrate on the website's design, development, and code,

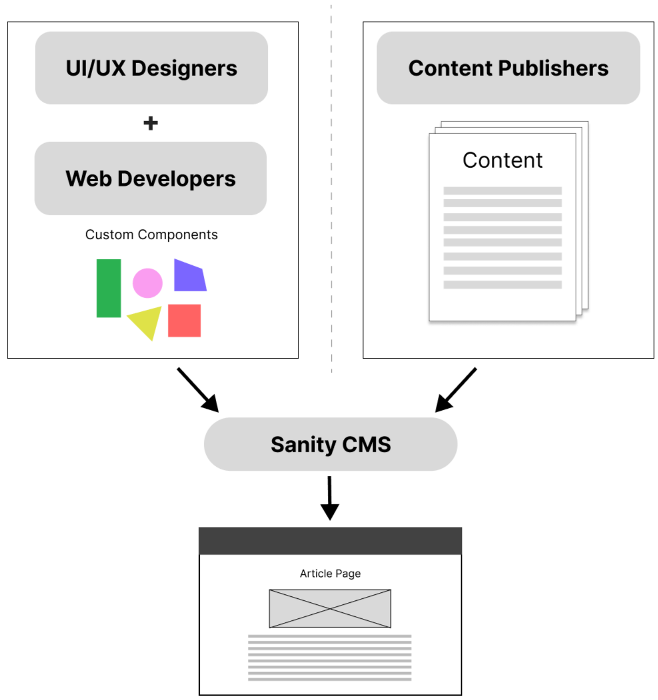

Figure 5: Sanity combines components and content for the frontend.

rather than being impeded by technicalities. Vercel is accessible because their documentation[6] for developers and management alike is easily found online.

They have a very generous free tier plan[7]. At this moment, we do not need the Pro Plan because the features are somewhat irrelevant to our needs. If an upgrade is ever needed in the future, I will notify those who may be of concern.

4.4 Worried About Certificates?

[6]https://vercel.com/docs
[7]https://vercel.com/pricing

5 Roles

I will build two teams: a design team and a tech team. The design team is responsible for creating the new UI/UX for the website. The tech team is responsible for code and technical utilities. The roles between each team is fluid. Those on the tech team can also work on the design team and vice versa. Members on each team will be selected and verified of their qualification for their prospective team by me, Neo Alabastro, and our current student director, Eva Weisenfeld.

I take the role of a delegate between three points of interest: technology, design, and accessibility. This project will not be the product of my own doing, but rather the collective effort of people with extensive skill sets in distinct disciplines. Therefore, my responsibility is to help them synthesize their work into a final product. In the case that I am unable to find certain people able to do certain types of specialized tasks, I will be responsible for that task's completion.

5.1 Recruiting

I will implement guerrilla recruiting techniques as well as more formal ones.

5.1.1 Design Team

IMA announcement

5.1.2 Development Team

ACM student chapter.

6 Project Timeline

The entire project consists of four phases: Repair, Design, Develop, and Teach.

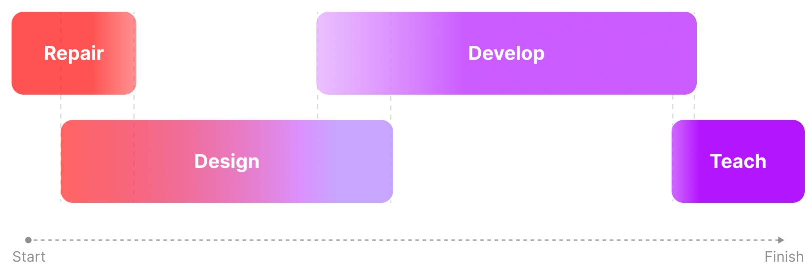

Figure 6: A simple graphic visualizing the phases.

The actual deadlines or sequential months are not given in figure 6. This should merely serve as a simplified timeline visualization. Note that these phases, although sequential, are also parallel. This shortens the time frame and design orientation, but requires more planning and organization.

6.1 Repair

This is a crucial phase that must happen as soon as possible. The goal of this phase is to fix the constant error that appears when visiting the site, as seen in figure 7.

For some context, I became interested in fixing the domain certificate issues back when OMS had a previous director by the name of Keigan, who I do not recall the last name of.

This is a backend issue, an issue with the website's certificate. An issue that should be addressed as soon as possible since it manifests as a problem in the frontend.

Moving to Cloudflare

Cloudflare has a step-by-step guide on their website detailing the steps and requirements to successfully switch domain registrars. In short, necessary credentials include:

- Credentials for GoDaddy
- Payment information to authorize domain payment
- Credentials for onmagnoliasquare@gmail.com

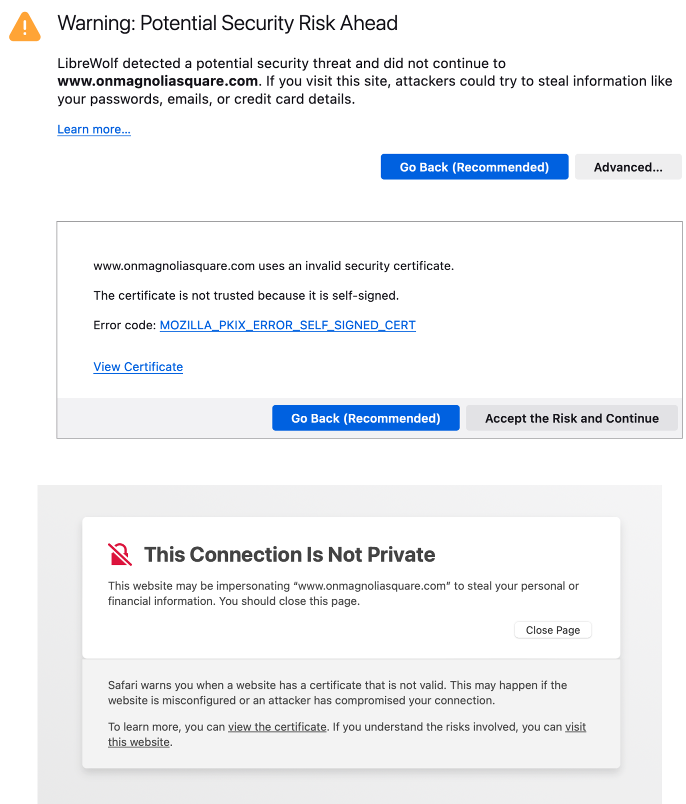

Figure 7: Flavors of the security warning, first in Firefox and next in Safari.

These items will enable me to successfully transition our DNS Registrar from GoDaddy to Cloudflare. Transferring a domain between registrars will not affect the domain name or TLD. Those will be exactly the same.

6.2 Design

This phase takes place during the fall semester of 2023, where we will make efforts to improve the design of the website. Design does not only include the way it looks, but also the way it feels and the way its used. This new design will be an overhaul of the current design and will feature better accessibility from desktop devices to mobile phones. Included in this process are the creation of usability interviews and studies, cross-platform coherent design system diagrams, and non-functional prototypes of the site itself.

6.3 Develop

During this phase, real world web development takes place, meaning a combination of asset and art creation, tech systems migration, and programming.

6.4 Teach

This final phase is about introducing and teaching the new website's features, functionality, and management. Teaching about these pivotal changes and functionality will occur during one or two team meetings. Due to the nature of a university student's social life and educational or non-educational commitments, I will personally reach out to those responsible for publishing content or maintaining the website in order to get them up to speed. Therefore, the meetings will be a general overview, and I will have personal meetings with respective OMS committee members for their benefit.

I will disseminate a reference document. One section of it will be for any future OMS developers who want to make changes to the website, and another section for any future designers who want to make changes to the website.

The developer section will contain explanations for infrastructure design decisions, common hiccups and issues one may encounter when making changes, and

Maintaining the infrastructure: issues of documentation and knowledge.

7 Conclusion

Parting people from their money is not a fun activity, but after reading this proposal, I hope that you will understand that this project's expenditure will not just be the work of one person, but the work of many people across different disciplines, and a collective effort of members of the NYUSH undergraduate community. That being said, the places where the money is spent is solely my responsibility. I will ensure it is spent in the right places and where it is most effective.

The entire point of this project is to get it off my hands; the maintenance of the website should be autonomous. I should not have to intervene with any issues after I graduate. However, such a high stake goal does not always manifest in the way you want it to. In the case that major issues arise that would perhaps require knowledged intervention, future OMS members are always welcome to email me in the future regarding how things work. I will be glad to teach them. Graduating from NYUSH only signifies my transition from a student to a mentor. I am always willing to give back to the community that created me.